

AI Document Intelligence Report

Executive Summary. Organizations ingest millions of PDFs annually across contracts, policies, research, and operations. Manual review does not scale. Document intelligence platforms combine layout-aware parsing, semantic chunking, vector retrieval, and large language models to answer questions with citations. This report outlines a reference architecture suitable for portfolio demonstrations on free-tier cloud infrastructure.

Problem Statement

Knowledge workers spend hours searching PDFs for clauses, metrics, and definitions. Keyword search fails on scanned tables and multi-column layouts. Generic chatbots hallucinate when not grounded. The gap is a system that retrieves evidence first, then generates answers constrained to retrieved passages. Success metrics include citation accuracy, latency under ten seconds, and clear separation between document-specific and general conversational queries.

Reference Architecture

Ingestion accepts PDF uploads per workspace. PyPDF or layout parsers extract text per page. Recursive character splitters produce overlapping chunks near nine hundred characters with one hundred twenty overlap. Embeddings use sentence-transformers/all-MiniLM-L6-v2. Vectors persist in per-document FAISS indexes on disk. The API layer is FastAPI with async lifespan hooks. Groq hosts llama-3.3-70b-versatile for generation with low temperature to reduce creativity on factual tasks.

Key Benefits

Grounded answers reduce hallucination risk in demo settings. Per-document indexes prevent cross-talk between unrelated files. Source previews let users verify page and chunk references. MMR retrieval balances relevance and diversity. The stack deploys on Hugging Face Spaces with secrets for API keys. Developers can fork the project and extend parsers, chunk sizes, or retrieval depth without redesign.

Limitations and Risks

OCR quality depends on source PDFs; scanned images may lose table structure. Chunk boundaries can split sentences and weaken retrieval for precise legal language. The demo does not provide professional legal, medical, or investment advice. Long documents may hit page caps. Cold starts on free tiers add latency for embedding model download. Concurrent uploads on ephemeral disks may not persist across Space restarts without persistent storage volumes.

Technologies Used

Backend: Python 3.11, FastAPI, Uvicorn. Orchestration: LangChain community integrations. Embeddings: Hugging Face sentence-transformers. Vector store: FAISS CPU. LLM: Groq ChatGroq. Frontend: React with per-document chat state in localStorage. Deployment: Hugging Face Spaces Docker SDK with app.py entrypoint and requirements.txt at repository root.

Security and Privacy

Demo deployments should not process regulated health or payment data without enterprise controls. API keys belong in Space secrets, never in client bundles. CORS restricts browser origins in production. Uploaded files reside on local disk; operators must understand data residency. Audit logs should record `document_id` and question length without storing full prompts if privacy is a concern.

Evaluation Methodology

Test questions should cover summarization, fact lookup, comparison, and negative cases where the answer is not in the document. Measure precision of cited pages, answer latency, and refusal behavior for general greetings. Regression suites can snapshot expected chunk_ids for canonical questions. Human reviewers score faithfulness on a five-point rubric for portfolio presentations.

Roadmap

Near-term: multi-document demo library, upload endpoint, delete with guardrails on default samples. Mid-term: hybrid search with BM25, table extraction, and streaming tokens to the UI. Long-term: role-based access, enterprise SSO, and batch indexing pipelines from object storage.

Conclusion

Document intelligence with RAG is a practical portfolio pattern. Isolated indexes per file, explicit citations, and honest limitation sections build trust with technical interviewers and product stakeholders.